

Domain 5 — Section Review



Context Management & Reliability

11/11

100% · Strong · 0 missed

Q1 ✓ **CORRECT**

Production monitoring reveals inconsistent synthesis quality. When aggregated results total ~75K tokens, the synthesis agent reliably cites information from the first 15K tokens (web search headlines and snippets) and the final 10K tokens (document analysis conclusions), but frequently omits critical findings that appear in the middle 50K tokens — even when those findings directly address the research question. How should you restructure the aggregated input?

- A** Implement rotation that alternates which subagent's results appear first across different research tasks, ensuring both sources receive primacy positioning equally over time.
- B** Place a key findings summary at the beginning of the aggregated input and organise detailed results with explicit section headers for easier navigation.
- C** Summarise all subagent outputs to under 20K tokens total before aggregation, ensuring content stays within the model's reliable processing range.
- D** Stream subagent results to the synthesis agent incrementally, processing web search results first to completion before introducing document analysis findings.

Why B is correct:

Lost-in-the-middle is structural — mitigate with position and structure, not rotation or size. Key findings summary at the beginning (reliable attention zone) + explicit section headers as structural anchors. Rotation (A) doesn't help — the problem is the *middle*, not which source comes first. Summarising to 20K (C) is lossy and destroys evidence from the middle. Streaming (D) misunderstands

the problem — it's about input structure, not delivery order.



Q2 ✓ **CORRECT**

After a 15-turn support session about a billing dispute, the agent's response refers to "a recent refund" and "prompt processing" — but the customer's original claim specified a \$247.83 refund promised by Friday. What context management failure caused this?

- A** The agent exceeded its token budget and silently dropped the tool result containing the refund amount.
- B** Progressive summarisation compressed specific transactional facts (\$247.83, by Friday) into vague prose during conversation history compression.
- C** The model's attention degraded due to the lost-in-the-middle effect applied to the conversation history.
- D** The tool result for the order lookup was too long and was truncated before reaching the model.

Why B is correct:

Progressive summarisation destroys high-value tokens first. "\$247.83 refund promised by Friday" becomes "a refund" and "promptly." Fix: extract transactional facts (amounts, dates, order IDs, customer-stated expectations) into a persistent 'case facts' block injected verbatim into every prompt, outside the summarised history. This block is never compressed.

Q3 ✓ **CORRECT**

A customer says: "I've been dealing with this for three weeks. Just connect me with a human — I'm done with bots." The agent responds: "I understand your frustration. Let me quickly check your account and see if I can resolve this directly." Is this correct behaviour?

A Yes — acknowledging frustration and attempting resolution maximises first-call resolution rate.



B Yes — the issue may be straightforward; the agent should try before escalating.

C No — explicit customer requests for a human agent must be honoured immediately without first attempting investigation.

D No — the agent should only escalate after a tool call confirms the issue exceeds its resolution capability.

Why C is correct:

Explicit human requests = immediate escalation, no exceptions. The customer stated their preference explicitly. Attempting investigation first violates that preference regardless of issue complexity. Correct escalation triggers: (1) customer explicitly requests human, (2) policy gap the agent can't fill, (3) inability to make meaningful progress. Sentiment alone is NOT a trigger — but an explicit request always is.

Q4 ✓ CORRECT

The document analysis agent discovers that two credible sources contain directly conflicting statistics on a key metric: one government report states 40% growth while an industry analysis states 12% growth. Both sources appear legitimate and the discrepancy could significantly affect the research conclusions. What's the most effective way for the document analysis agent to handle this?

A Select the government report figure as the more authoritative source and note the discrepancy briefly.

B Average the two values (26%) and include a note that sources vary on this metric.

C Flag the conflict to the coordinator and pause document analysis pending coordinator

resolution.



- D** Include both statistics with full source attribution and explicitly annotate the conflict, letting the synthesis agent and downstream consumers decide how to interpret the discrepancy.

Why D is correct:

Surface conflicts with both values and attribution. Never average. Never silently pick. Both sources are credible. The document analysis agent should not arbitrate — it should preserve both figures with full source attribution and explicitly annotate the conflict. Averaging (B) fabricates a number neither source reported. Pausing (C) blocks progress unnecessarily — the coordinator can handle this information during synthesis.

Q5 ✓ **CORRECT**

The web search subagent returns results for only 3 of 5 requested source categories (competitor websites and industry reports succeeded, but news archives and social media feeds timed out). The document analysis subagent successfully processed all provided documents. The synthesis subagent must now produce a findings summary from this mixed-quality input. What's the most effective error propagation strategy?

- A** Halt synthesis and retry all failed searches with extended timeouts before proceeding.
- B** Proceed with available results and exclude any sections in the synthesis where data is missing.
- C** Structure the synthesis output with coverage annotations indicating which areas are well-supported versus which have gaps due to the unavailable source categories.
- D** Have the coordinator re-invoke the web search agent for only the failed categories before proceeding to synthesis.

Why C is correct:

Graceful degradation with coverage annotations — not workflow termination. Partial results from succeeded subagents should not be discarded when one subagent fails. The synthesis should proceed using available data and explicitly annotate which findings are well-supported versus which topic areas have coverage gaps due to the unavailable sources. This preserves the value of succeeded work while maintaining transparency about limitations.

Q6 ✓ **CORRECT**

During extended codebase exploration (context is 82% full), the model starts referencing 'typical Spring MVC patterns' instead of the specific class hierarchy it identified 20 turns ago. Recommended immediate action:

- A** Start a new session with an injected structured summary of key findings discovered so far.
- B** Switch to a model with a larger context window to accommodate the full session.
- C** Use `/compact` to reduce context usage while preserving key findings, then continue the session.
- D** Ask the model to re-read the specific files it analysed earlier to refresh its memory.

Why C is correct:

/compact for full-but-valid context; fresh start only for stale context. Context is 82% full but valid — the specific class hierarchy was found. `/compact` compresses verbose intermediate steps while preserving key findings, letting the session continue without losing recent discoveries. A fresh start (A) would lose all valid context gathered so far. Re-reading files (D) would exhaust the remaining 18% context budget.

Q7 ✓ CORRECT



Research subagents return verbose reasoning chains and full source text to the coordinator. The coordinator is hitting context limits and synthesis quality is degrading. Most effective fix:

- A Increase the coordinator's max_tokens parameter to accommodate larger inputs.
- B Instruct the coordinator to summarise incoming subagent results before processing them.
- C Modify subagents to return structured data (key facts, citations, relevance scores) instead of verbose prose and reasoning chains.**
- D Reduce the number of subagents to lower total output volume.

Why C is correct:

Fix context bloat upstream, not downstream. Reasoning happens inside each subagent's context — only distilled conclusions (key facts, citations, relevance scores) should cross agent boundaries. Having the coordinator summarise (B) adds processing without fixing the source. max_tokens (A) is a band-aid. Fewer subagents (D) degrades coverage. The cheapest effective fix is always modifying what subagents output.

Q8 ✓ CORRECT

During a materials research task, the web search subagent queries three source categories with different outcomes: academic databases returned 15 relevant papers, industry reports returned '0 results found,' and patent databases returned 'Connection timeout.' When designing error propagation to the coordinator, what approach enables the best recovery decisions?

- A Return a single combined error status for all three categories, since one of them failed.
- B Return only the successful academic database results and silently discard the other

outcomes.



- C** Return all three outcomes separately: academic results as success, industry as `isError:false` with empty results (valid query, no data), and patent as `isError:true` with structured error context including failure type, query attempted, and retry guidance.
- D** Automatically retry the patent database timeout within the subagent, returning a unified result once all categories are complete or exhausted.

Why C is correct:

Distinguish access failure from valid empty result from success — each requires a different coordinator response. Academic = success (15 papers, proceed). Industry = `isError:false` + empty results (looked and found nothing — valid, no retry needed). Patent = `isError:true` with structured context (connection timeout — may be worth retrying with a different query or at a later time). Conflating these (A, B) destroys the coordinator's ability to make targeted recovery decisions. Silent retry (D) may resolve the timeout but hides the failure context that helps the coordinator decide whether broader retry or fallback is needed.

Q9 ✓ **CORRECT**

You're running an extended codebase exploration session to map a legacy system's data flow. After 30+ turns, the context is 78% full and Claude has identified key service boundaries and data transformations. You need to continue across multiple future sessions without losing these findings. What's the most reliable approach?

- A** Use `/compact` to compress the current session and continue from where you left off.
- B** Ask Claude to summarize its findings in a response, then manually copy them into a document.
- C** Have Claude write the key findings — service boundaries, data flow maps, identified dependencies — to a scratchpad file in the repo that future sessions can read.

- D Start a new session with a detailed prompt describing the codebase from me



Why C is correct:

Scratchpad files persist findings across context boundaries and session restarts. Claude writes structured findings to a file; future sessions load it and continue with full context. /compact (A) extends the current session but doesn't survive a crash or persist to a new session. Summarizing in a response (B) requires manual effort and loses structure. Memory prompts (D) are lossy and depend on human recall — they don't preserve the specific findings Claude discovered.

Q10 ✓ **CORRECT**

Your extraction pipeline reports 96% accuracy across 10,000 invoices processed last month. Your team proposes removing human review entirely based on this metric. What validation step is most critical before automating the full pipeline?

- A Run the pipeline on 1,000 additional invoices to confirm the 96% figure is stable at scale.
- B Analyze accuracy by document type and field segment to verify consistent performance before removing human review.**
- C Lower the confidence threshold from 0.9 to 0.85 to approve more extractions automatically and increase throughput.
- D Add a second extraction pass to catch the 4% of errors the first pass misses.

Why B is correct:

Aggregate accuracy masks catastrophic error rates on specific segments. A pipeline that's 99.9% accurate on standard digital invoices but 40% accurate on handwritten or scanned ones still reports ~96% overall. Before removing human review, validate accuracy by document type, field, and format. Confidence threshold adjustment (C) and second passes (D) are optimisations — neither validates

whether the 96% is evenly distributed. Additional volume (A) confirms stability, not consistency across segments.



Q11 ✓ **CORRECT**

A customer contacts support saying 'I need help with my account.' Your agent calls `get_customer` and the tool returns three different customer records matching that name. What should the agent do?

- A Select the most recently active account — it is most likely the customer contacting support.
- B Select the account with the highest lifetime value to prioritize high-value customers.
- C Ask the customer for an additional identifier (email address, account number, or order number) to resolve the ambiguity before proceeding.**
- D Escalate immediately to a human agent since the agent cannot determine the correct account without additional information.

Why C is correct:

Multiple matches → request additional identifiers. Never select by heuristic. Heuristic selection (A, B) risks acting on the wrong account — potentially exposing one customer's data to another or making unauthorized changes. The agent can resolve this itself by asking a follow-up question: email, account number, or recent order ID. Immediate escalation (D) is unnecessarily disruptive when a clarifying question resolves the ambiguity.

See Final Results →

